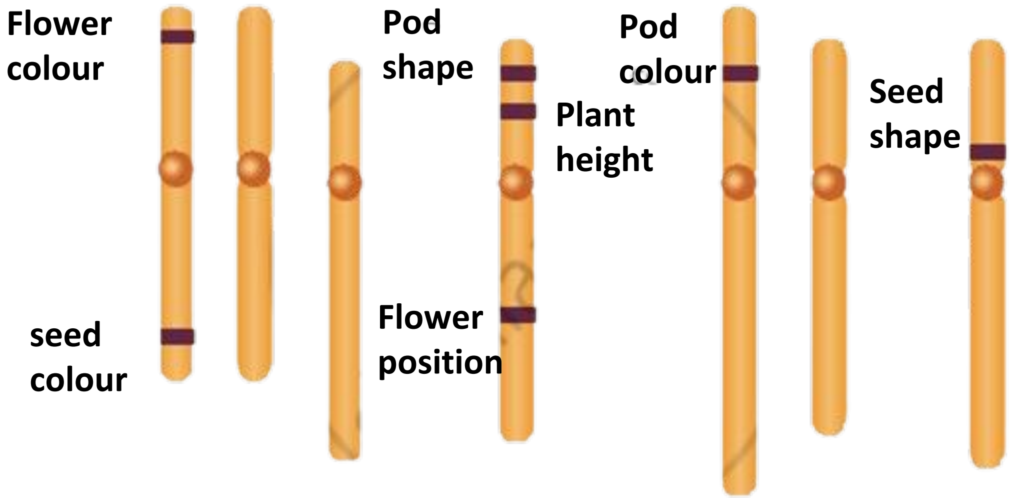
# Linkage

*Dr. Ayan Sadhukhan*
*BSBE, IIT Jodhpur*




**Mendel's law of independent assortment of pairs of characters breaks down if the 2 genes are linked, i.e physically close on the chromosome**

Mendel had extremely good fortune in his classic experiments with the garden pea.

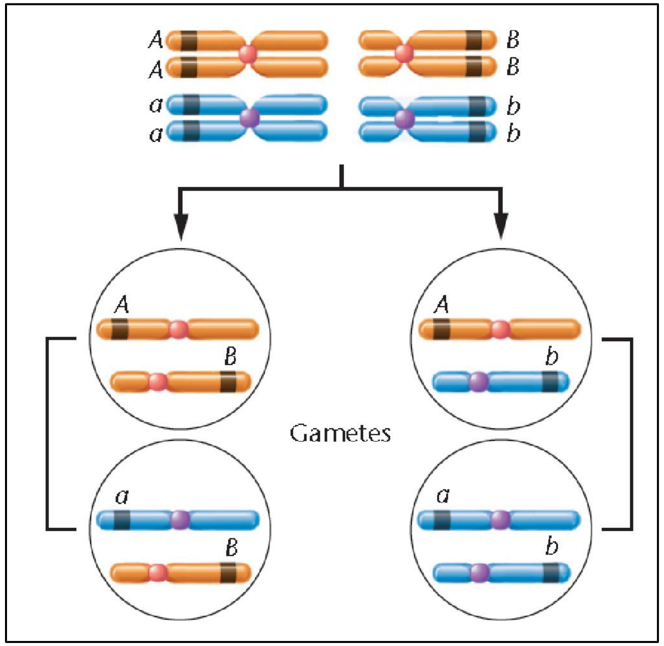
He did not encounter any linkage relationships between the seven mutant characters in his crosses.

Had Mendel obtained highly variable data characteristic of linkage and crossing over, he might not have succeeded in recognizing the basic patterns of inheritance and interpreting them correctly.

# Linkage

## Independent assortment

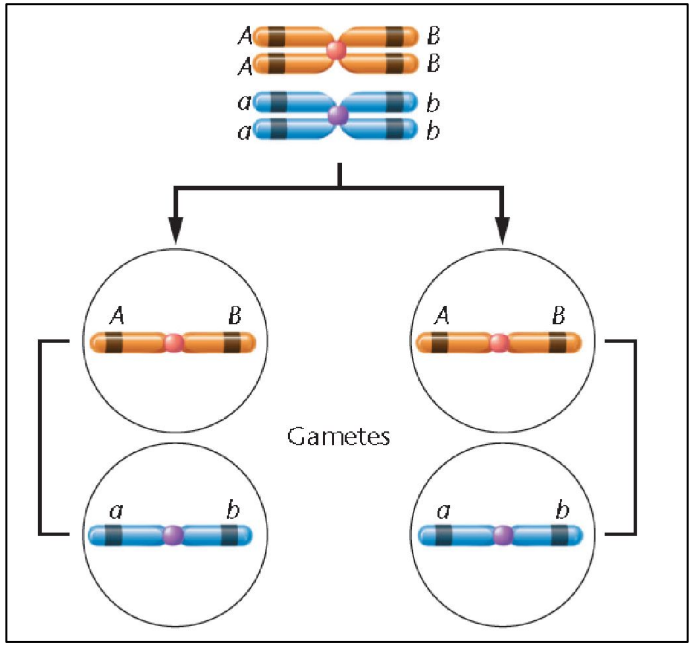
**Two genes on two different homologous pairs of chromosomes**




## Linkage without crossing over

**Two genes on single homologous pair of chromosomes – no crossover occurs**

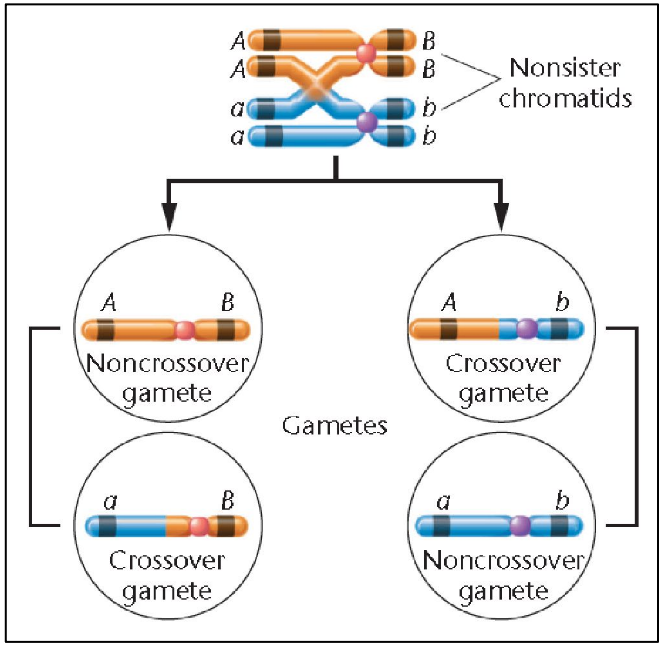
*complete linkage* (**<u>theoretical</u>**)




*parental/ non-crossover gametes*

## Linkage with crossing over

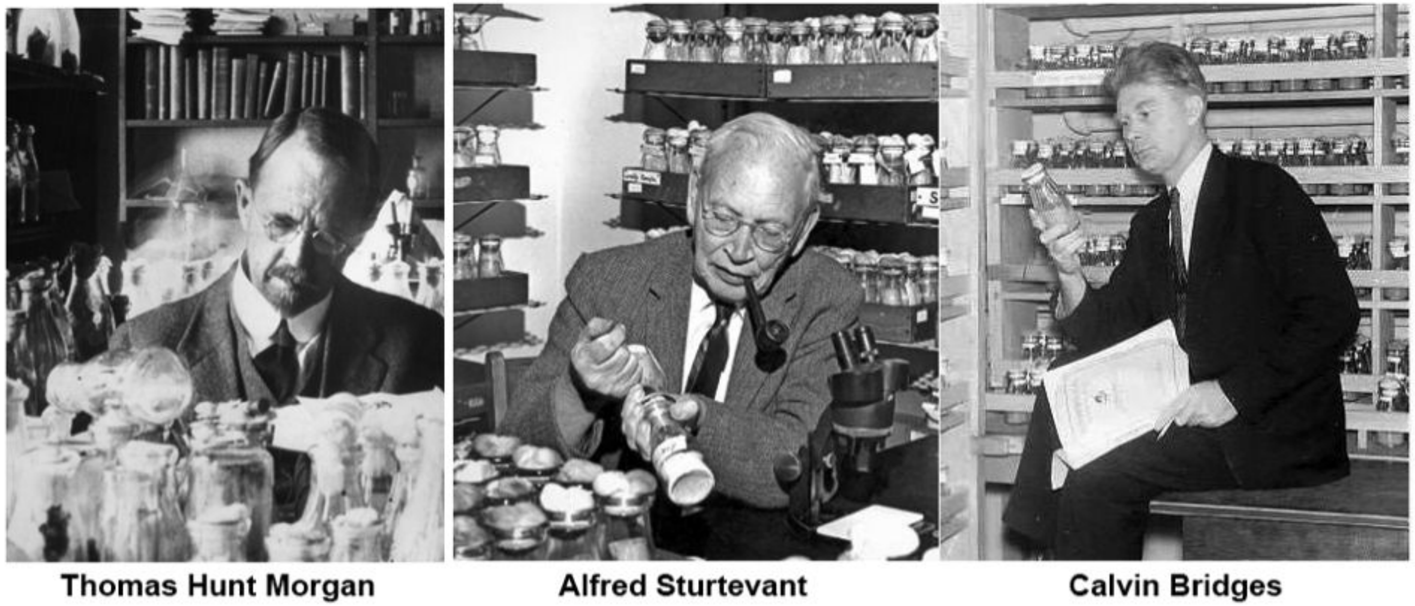
**Two genes on single homologous pair of chromosomes – crossover occurs**



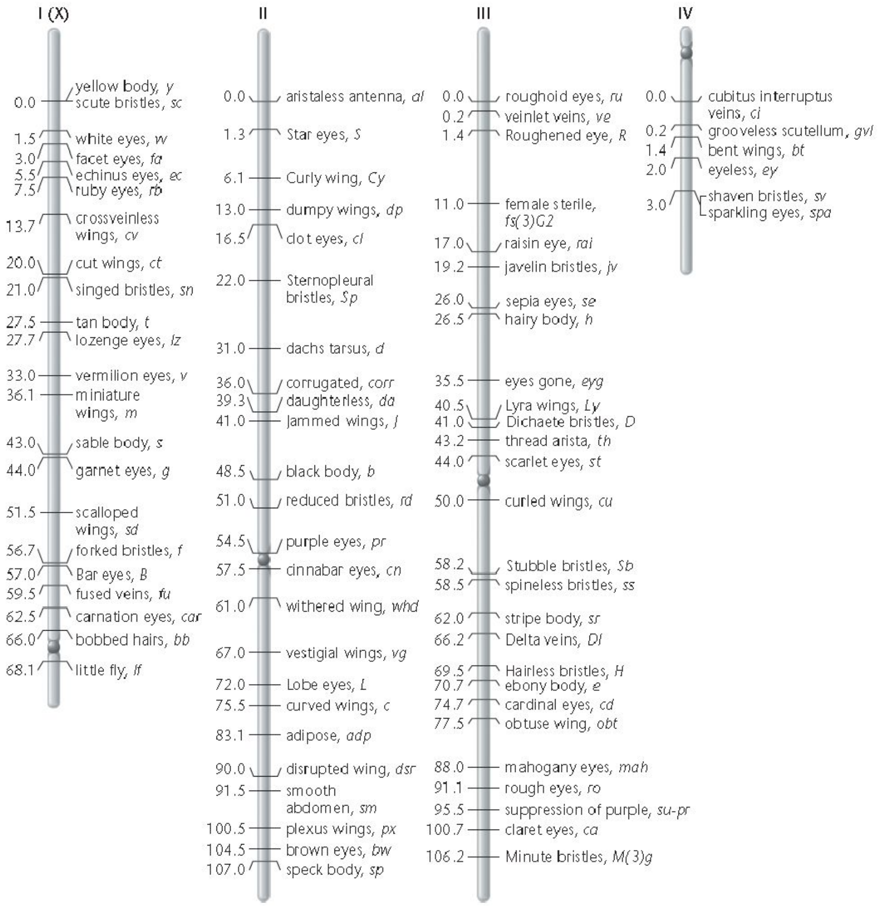

*Recombinant/ crossover gametes*

**the frequency with which crossing over occurs between any two linked genes is proportional to the distance separating the respective loci along the chromosome. As the distance between the two genes increases, the proportion of recombinant gametes increases and that of the parental gametes decreases**

# Morgan and Sturtevant's crosses with Drosophila led to the discovery of crossover – the basis of gene mapping in eukaryotes

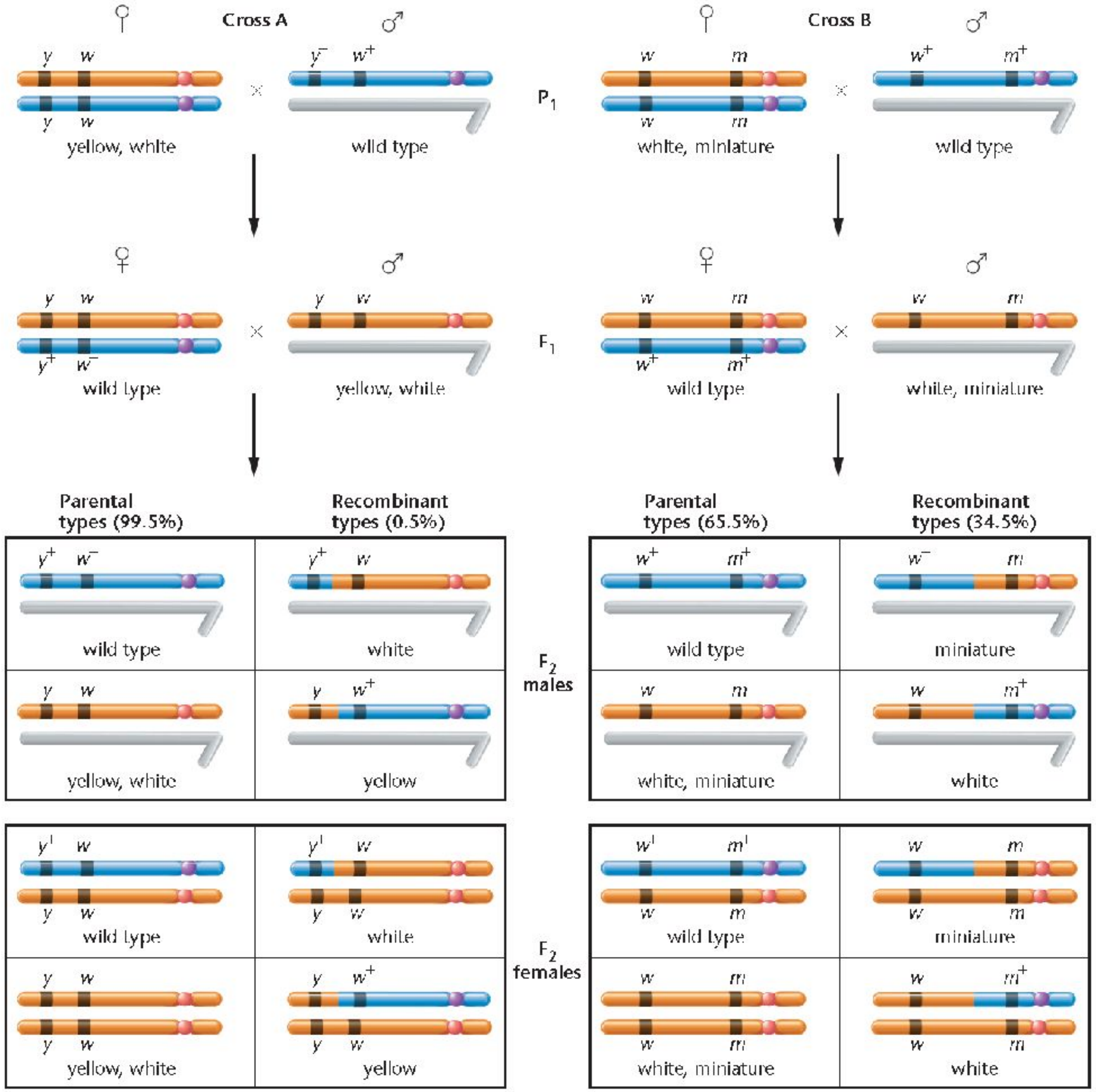



**The bottles they are holding contained these flies and their mutants**




**Gene map of the Drosophila genome**

# How they discovered crossing over ?




Morgan was the first to discover the phenomenon of X-linkage. In his studies, he investigated numerous *Drosophila* mutations located on the X chromosome.

He was fine when he used only one gene at a time on the X chromosome.

But **his results were puzzling** he made crosses **involving two X-linked genes**, his results were initially puzzling

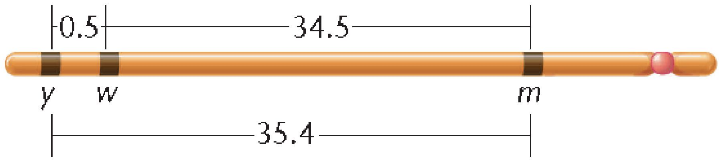
female flies expressing the mutant ***yellow* body (*y*)** and ***white* eyes (*w*)** alleles were crossed with **wild-type** males (**gray body and red eyes** ).

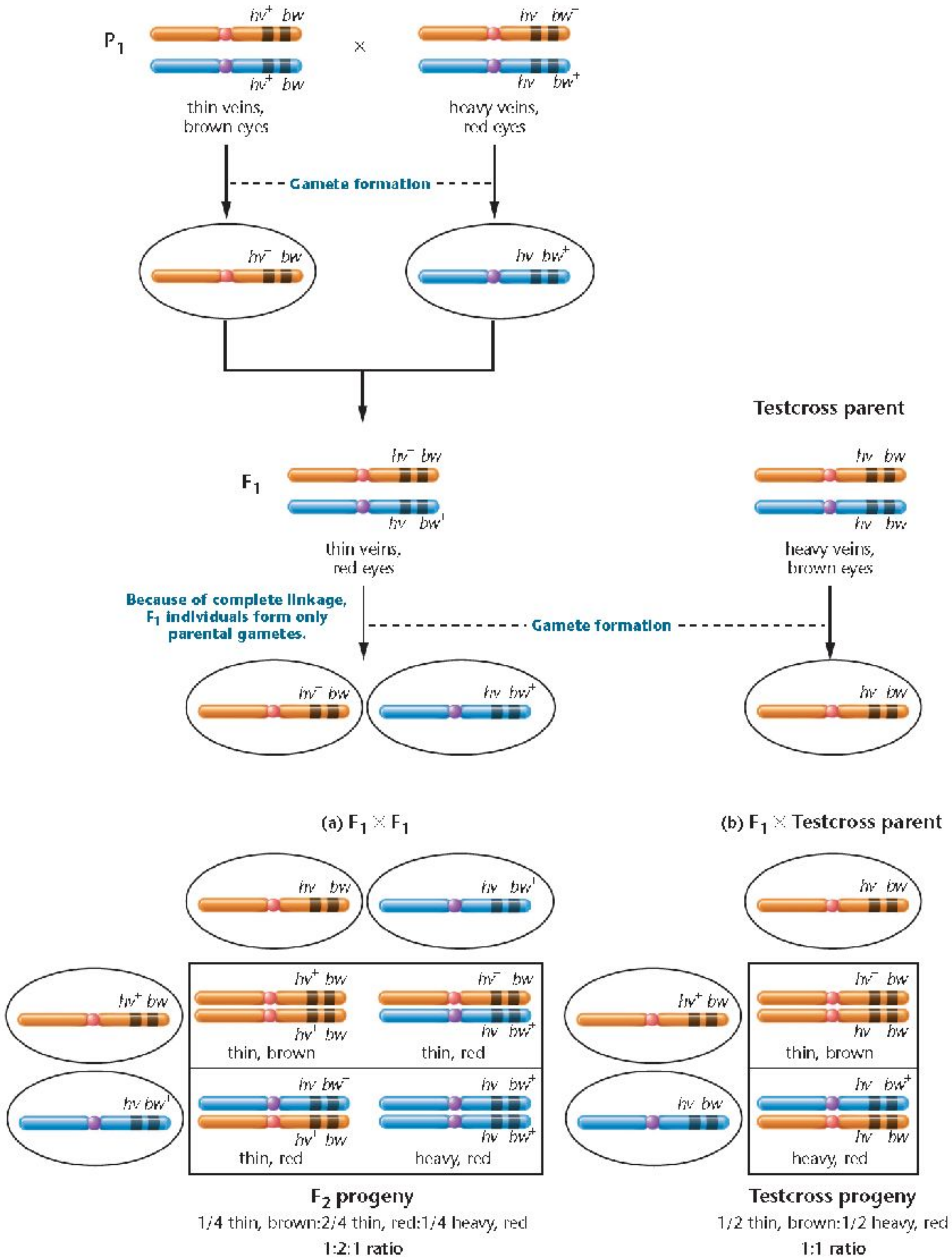
The F1 females were wild type, while the F1 males expressed both mutant traits.

In the F2 the vast majority of the total offspring showed the expected **parental phenotypes —yellow-bodied, white eyed** flies and wild-type flies (**gray-bodied, red-eyed** ).

The remaining flies, less than 1.0 percent, were either **yellow-bodied with red eyes** or **gray-bodied with white eyes** .

It was **as if the two mutant alleles had somehow separated from each other** on the homolog during gamete formation in the F1 female flies.



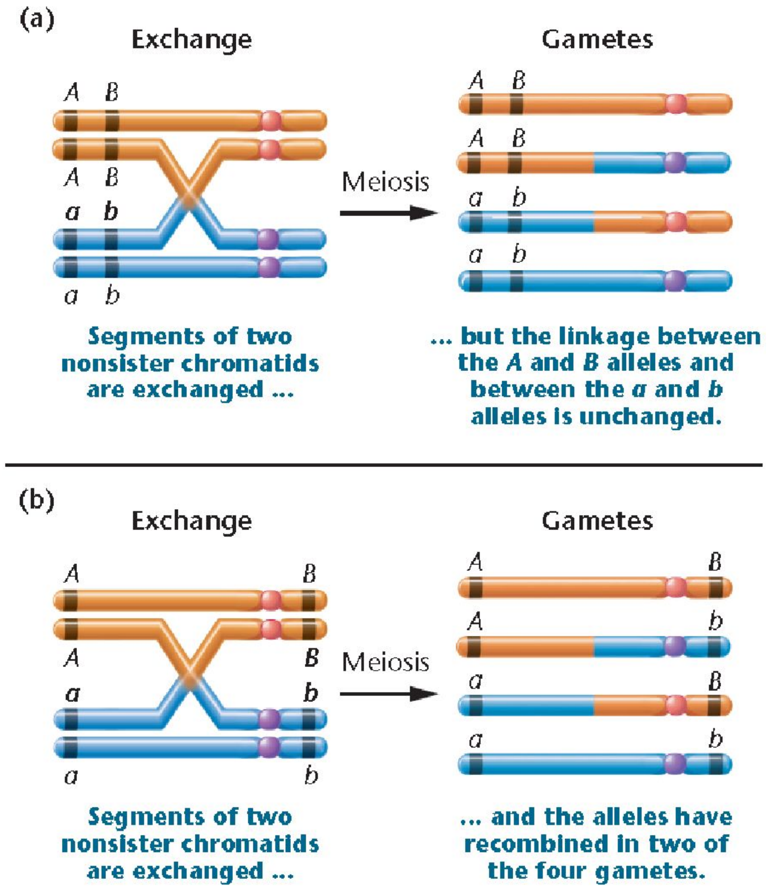



# Complete Linkage

### Linkage ratio

If complete linkage exists between two genes because of their close proximity, and organisms heterozygous at both loci are mated, a unique F2 phenotypic ratio (1:2:1) results

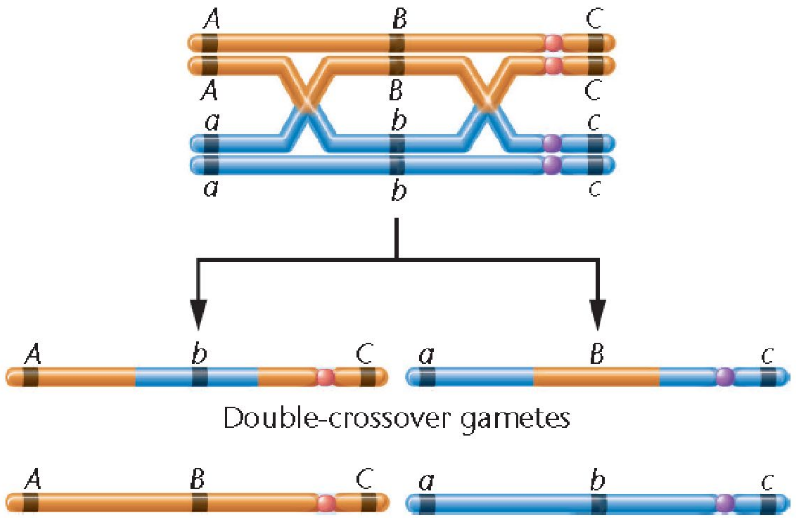
# Single crossover




(a) the exchange does not alter the linkage arrangement between the alleles of the two genes, only parental gametes are formed, and the exchange goes undetected.

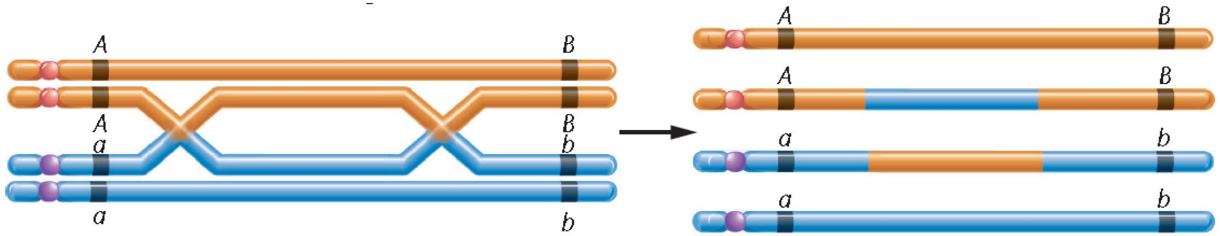
In (b) the exchange separates the alleles, resulting in recombinant gametes, which are detectable.

# Double crossover



Double-crossover gametes

Noncrossover gametes




No detectable recombinants

Double crossover events may change the allele order on a chromosome

# Three-point mapping problem

3 mutant traits

*yellow* body color, *echinus* (rough) eye shape, *white* eye color

*WT*
Grey body, red eye color, normal eye shape

*y* *ec* *w*

$y^+$ $ec^+$ $w^+$

Parents


×

♀ ♂

F1 all $F_1$ females are wild type,
while all $F_1$ males are yellow, white, and echinus

selfing

| F2 phenotypes | Observed number | Percentage | |
|---|---|---|---|
| y ec w<br>$y^+$ $ec^+$ $w^+$ | 4685<br>4759 | 94.44% | Non crossover |
| y $ec^+$ $w^+$<br>$y^+$ ec w | 80<br>70 | 1.5% | Single crossover |
| y $ec^+$ w<br>$y^+$ ec $w^+$ | 193<br>207 | 4% | |
| y ec $w^+$<br>$y^+$ $ec^+$ w | 3<br>3 | 0.06% | Double crossover |

**this order (*y ec w*) may be incorrect**

**What is the correct order of the 3 genes on the chromosome?**

# How to determine the correct order of the 3 genes from the F2 results?

Check change in the order of parental alleles

| F2 phenotypes | Observed number | Percentage |
|---|---|---|
| y ec w<br>$y^+$ $ec^+$ $w^+$ | 4685<br>4759 | 94.44% |
| *y* $ec^+$ $w^+$<br>$y^+$ *ec* *w* | 80<br>70 | 1.5% |
| *y* $ec^+$ *w*<br>$y^+$ *ec* $w^+$ | 193<br>207 | 4% |
| *y* *ec* $w^+$<br>$y^+$ $ec^+$ *w* | 3<br>3 | 0.06% |

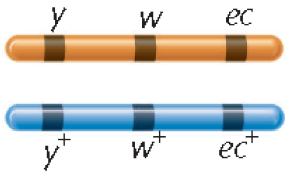
Non crossover (parental)

Single crossover between *w* and *ec*

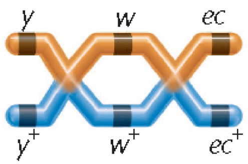
Double crossover between *y* and *w* and between *w* and *ec*

Observe between which 2 genes there are exchanges

If we assume this order is correct



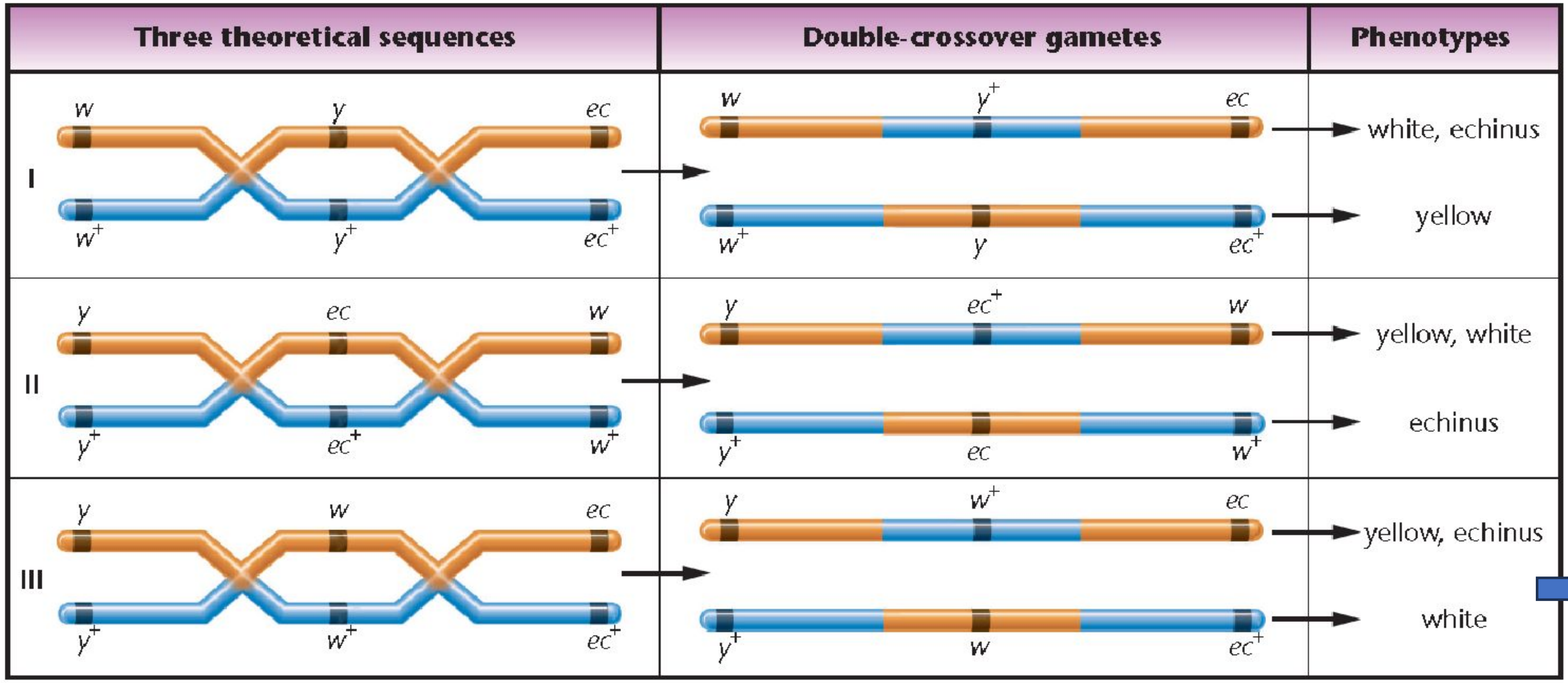

Parental




Double crossover

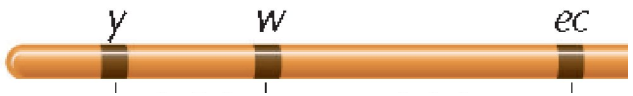
It makes sense
(*w* must be in the center)

# Let’s explore all 3 possibilities




This types of DCO offspring were actually observed.

So, this order is correct



# Calculation of distances

***y – w* distance**

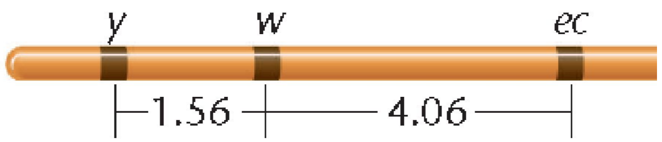
- DCO between *y* and *w* = 3+3= 6
- SCO between *y* and *w* = 80+70= 150
- Total Crossovers between *y* and *w* = 156
- 156/10000*100= 1.56 map units

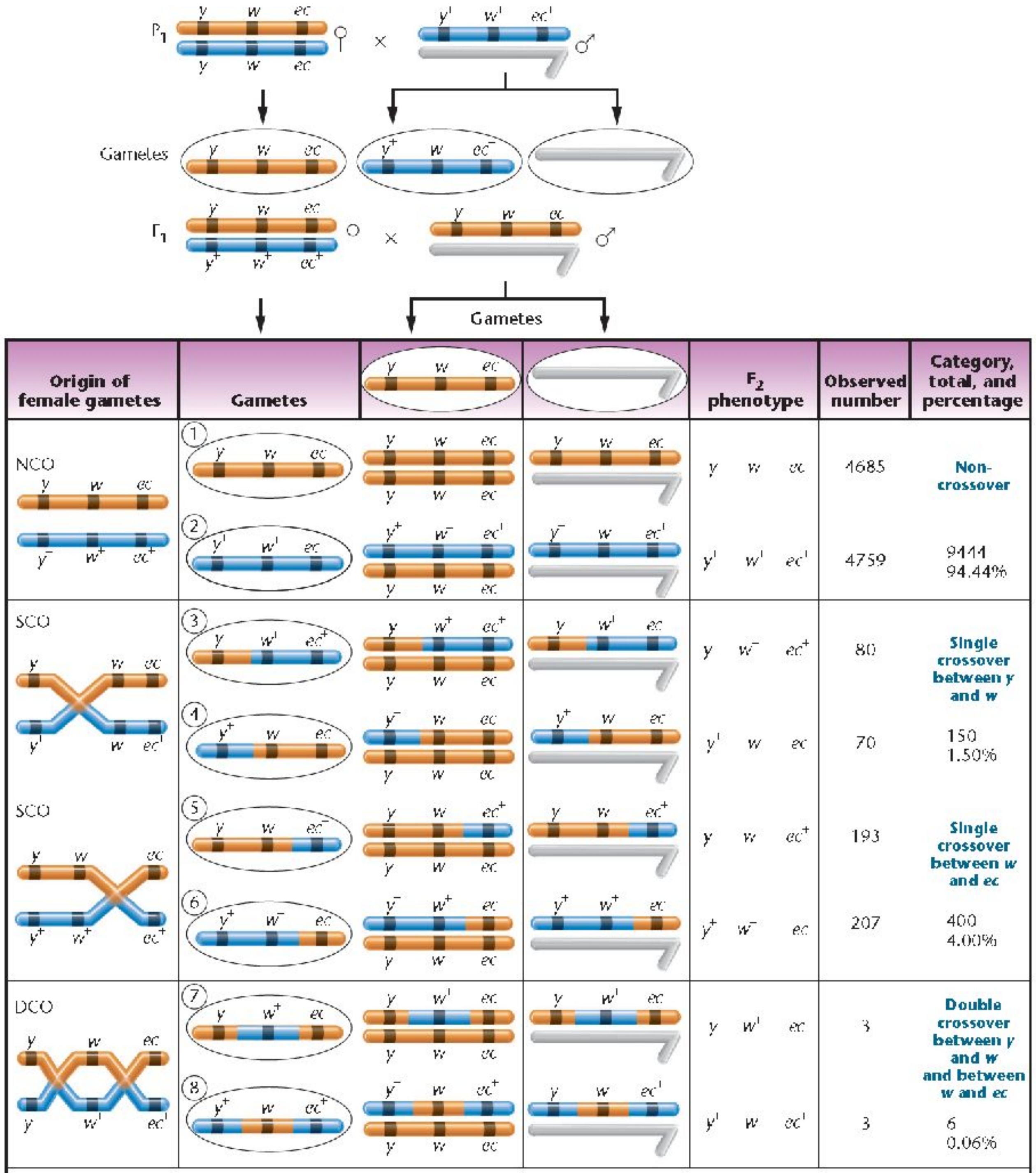
***w - ec* distance**

- DCO between *w* and *ec* = 3+3= 6
- SCO between *w* and *ec* = 193+207=400
- Total Crossovers between *w* and *ec* = 406
- 406/10000*100= 4.06 map units

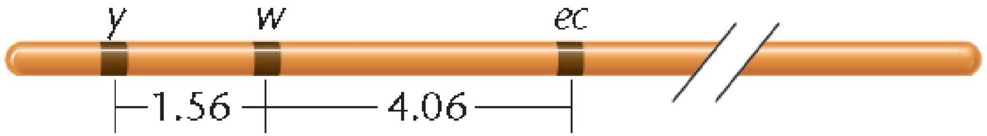
| F2 phenotypes | Observed number | Percentage |
|---|---|---|
| y w ec<br>$y^+$ $w^+$ $ec^+$ | 4685<br>4759 | 94.44% |
| $y$ $w^+$ $ec^+$<br>$y^+$ $w$ $ec$ | 80<br>70 | 1.5% |
| $y$ $w$ $ec^+$<br>$y^+$ $w^+$ $ec$ | 193<br>207 | 4% |
| $y$ $w^+$ $ec$<br>$y^+$ $w$ $ec^+$ | 3<br>3 | 0.06% |

Total 10000






# Three-point mapping problem

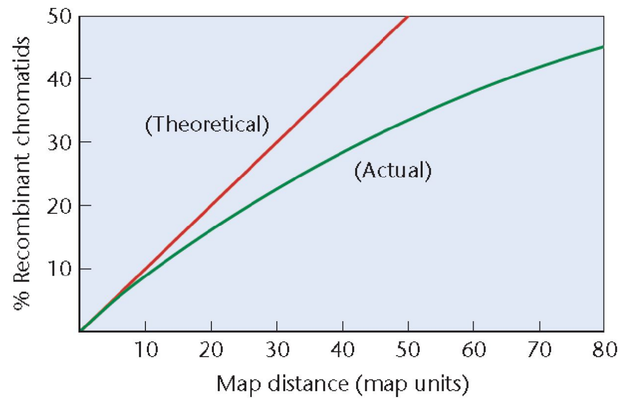


# Interference and coefficient of coincidence

$DCO_{exp}$ = product of the two single crossovers that make up a double crossover occurring independently of one another (product law of probability)

$DCO_{obs}$ = observed ratio of the DCO offspring

$DCO_{exp} = (0.015) * (0.04) = 0.0006 = 0.06\%$



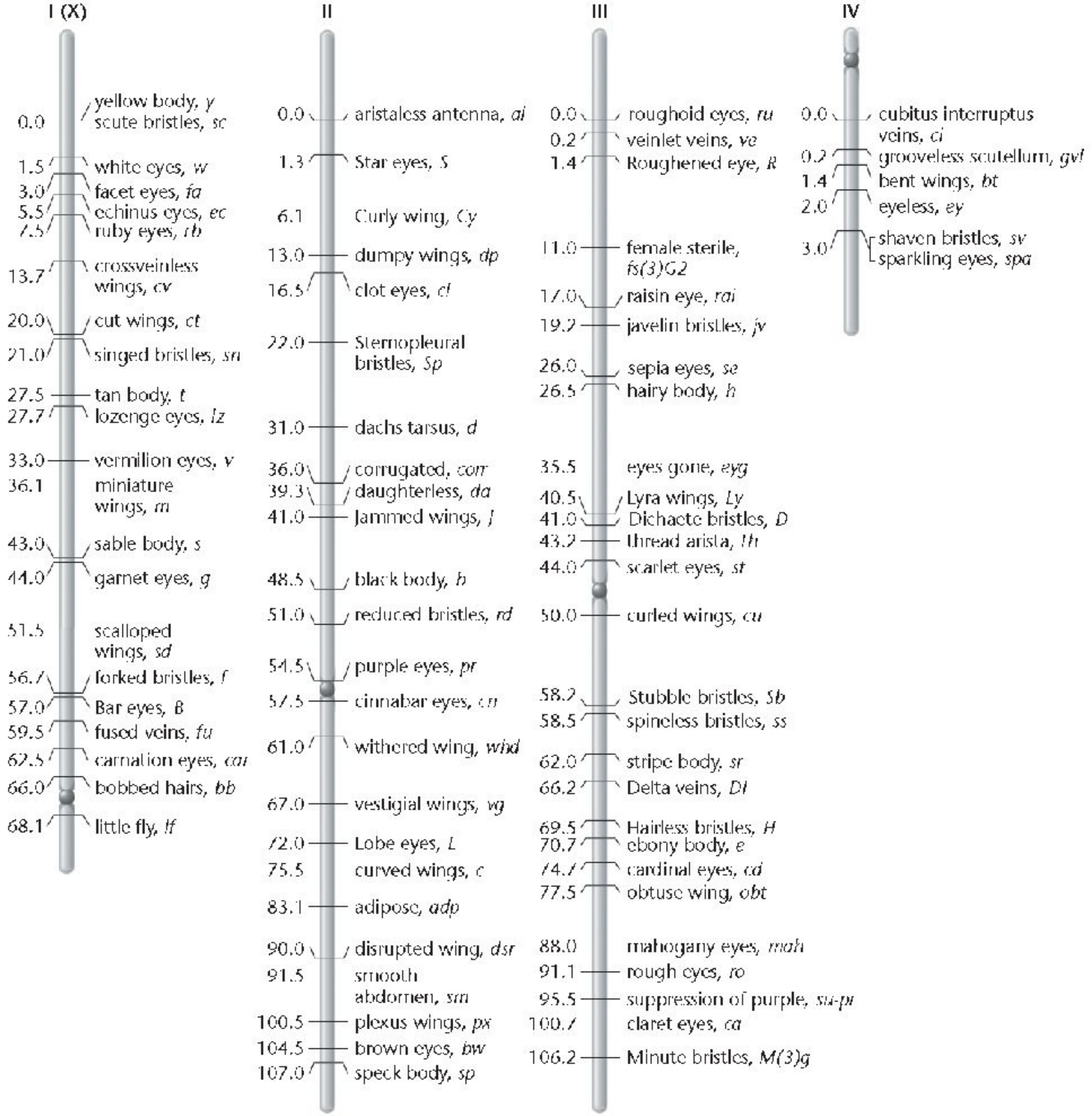

**The observed DCO is less than the expected DCO**

**This is due to**

**Interference (*I*)**, the inhibition of further crossover events by a crossover event in a nearby region of the chromosome

**coefficient of coincidence** $$C = \frac{\text{Observed DCO}}{\text{Expected DCO}}$$

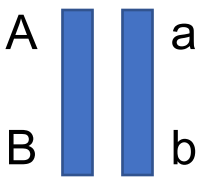
$$I = 1 - C$$



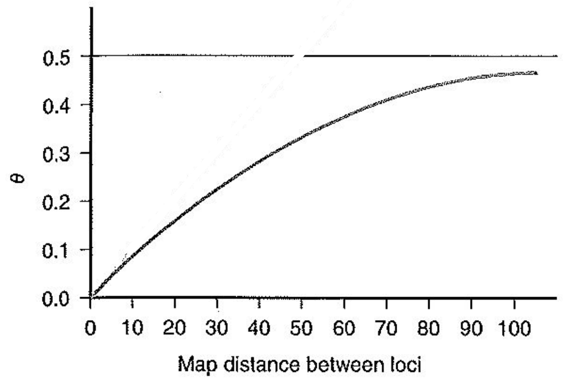

# The complete map of the Drosophila genome

# Recombination frequency

$$\Theta = \frac{\text{Total amount of recombinants}}{\text{Total amount of recombinants + Total amount of non-recombinants}}$$




| Parent | Gametes | Theta |
|---|---|---|
| | 50% non-rec and 50% rec | 0.5 |
| | 90% non-rec and 10% rec | 0.1 |
| | 99% non-rec and 1% rec | 0.01 |
| | 100% non-rec | 0 |



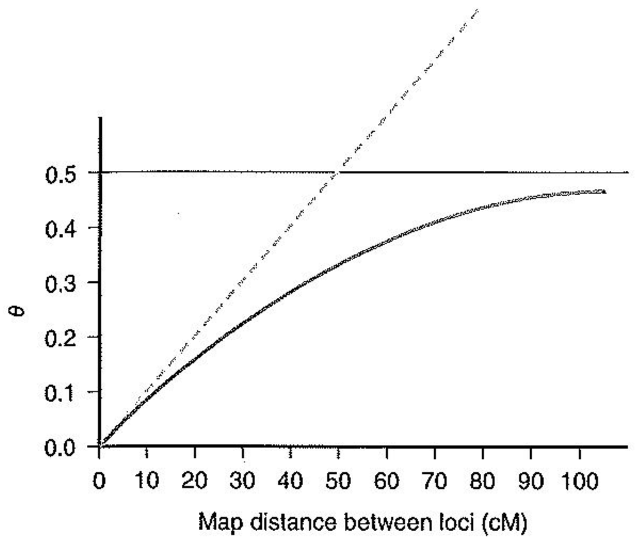
# Genetic distance

Genetic distance = the genetic length over which one crossover occurs in 1% of meiosis. This distance is expressed in cMorgan.

***1 cMorgan = 0.01 recombinants = average of 1Mb (physical distance)***

(Assuming that the recombination frequency is uniform along the chromosomes)

As double recombinants occur the further two loci are, the frequency of recombination does not increase proportionately.



# LOD score method for Linkage analysis

The difficulty arises when two genes of interest are separated on a chromosome such that recombinant gametes are formed, obscuring linkage in a pedigree. In such cases, an approach relying on probability calculations, called the lod score method, helps to demonstrate linkage. First devised by J. B.S. Haldane and C. A. Smith in 1947 and refined by Newton Morton in 1955, the **lod score** (standing for **log of the odds favoring linkage**) **assesses the probability that a particular pedigree involving two traits reflects linkage.**

First, the probability is calculated that family data (pedigrees) concerning two or more traits conform to the transmission of traits without linkage.

Then the probability is calculated that the identical family data following these same traits result from linkage with a specified recombination frequency.

The ratio of these probability values expresses the "odds" for, and against, linkage.
The lod score method represented an important advance in assigning human gene to specific chromosomes and in constructing preliminary human chromosome maps. However, its accuracy is limited by the extent of the pedigree, and the initial results were discouraging because of these limitations and because of the relatively high haploid number of human chromosomes (23).

Linkage analysis is a method that is used to decide if two loci or a loci and a disease gene are linked :

1. Ascertain whether the recombination fraction theta between two loci deviates significantly from 0.5.

2. If theta is different from 0.5, we need to make the best estimate of theta, since this parameter tells us how close the linked loci are.

Linkage is expressed as a LOD score (Z); a *" logarithm of odds"*

$$\text{LOD score } (\Theta) = \log_{10} \frac{\text{Likelihood of linkage}}{\text{Likelihood that loci are unlinked (theta = 0.5)}}$$

Positive values of Z at a given Θ suggest that two loci are linked.

Negative values of Z at a given Θ suggest that two loci are not linked.

By convention, a LOD score of +3 or greater is considered definitive evidence that two loci are linked. A LOD score below -2 excludes linkage.

$$\text{Log} \frac{1000}{1}$$

Probability of a recombination is Θ
N is amount of **recombinants** in pedigree

Probability that no recombination will occur is (1-Θ)
M is amount of **non-recombinants** in pedigree

$$Z(\Theta) = \log_{10} \frac{\Theta^N}{(0.5)^N} + \log_{10} \frac{(1-\Theta)^M}{(0.5)^M} = \log_{10} \frac{\Theta^N (1-\Theta)^M}{(0.5)^N \ (0.5)^M}$$

# Chromosome Mapping Is Now Possible Using DNA Markers and Annotated Computer Databases

Traditional methods of mapping - based on recombination analysis

maps in other organisms (including humans) – are based on other methods

**DNA markers** are short segments of DNA whose sequence and location are known, making them useful landmarks for mapping purposes  study genes in relation to markers has extended our knowledge of the location within the genome of countless genes

Association of markers to phenotypes  **association mapping**  lead to the discovery of causative genes

DNA Markers –
**restriction fragment length polymorphisms (RFLPs)**
RFLPS are polymorphic sites generated when specific DNA sequences are recognized and cut by restriction enzymes.

**microsatellites** or **short tandem repeats**
Microsatellites are short repetitive sequences that are found throughout the genome, and they vary in the number of repeats at any given site. For example, the two-nucleotide sequence CA is repeated 5–50 times per site [(CA)n] and appears throughout the genome approximately every 10,000 bases, on average. Microsatellites may be identified not only by the number of repeats but by the DNA sequences that flank them.

**single-nucleotide polymorphisms** (**SNPs)**
More recently, variation in single nucleotides, called **single-nucleotide polymorphisms** (**SNPs**, also called “snips”), has been utilized. Found throughout the genome, up to several million of these variations may be screened for an association with a disease or trait of interest, thus providing geneticists with a means to identify and locate related genes.

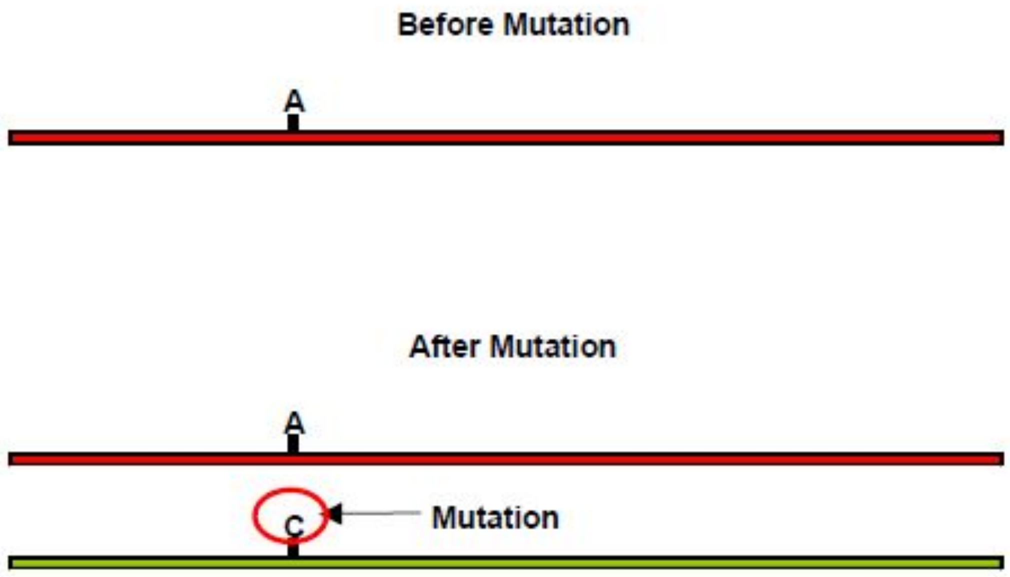
**What is linkage disequilibrium?**

**Linkage equilibrium** occurs when the genotype present at one locus is independent of the genotype at a second locus**.**

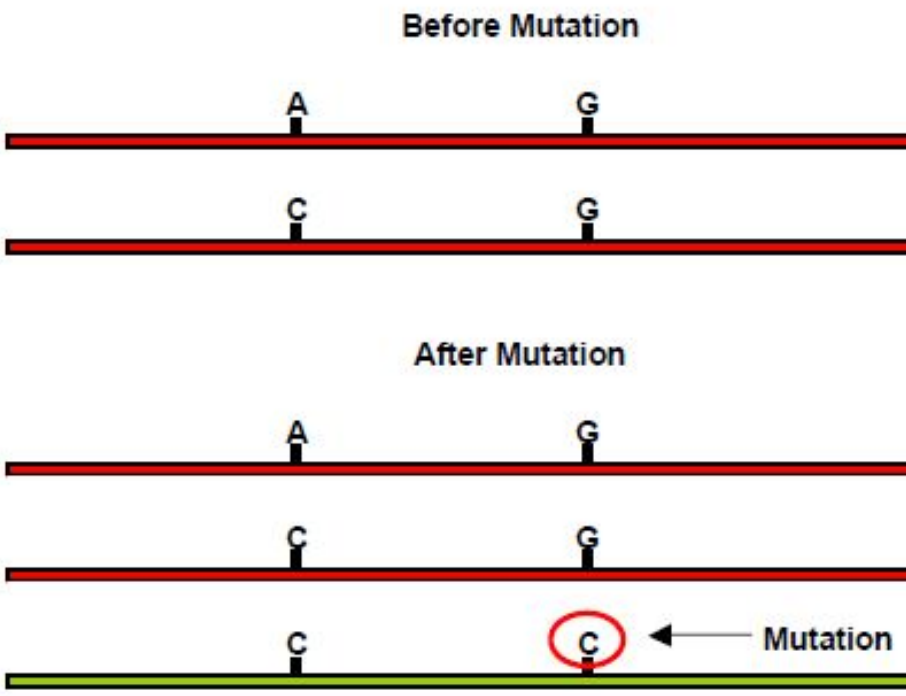
**Linkage disequilibrium** occurs when genotypes at the two loci are not independent of another.

**Linkage disequilibrium (LD)**
Measures the degree to which alleles at two loci are associated
o The non-random associations between alleles at two loci
 Based on expectations relative to allele frequencies at two loci

# Alleles that exist today arose through ancient mutation events…



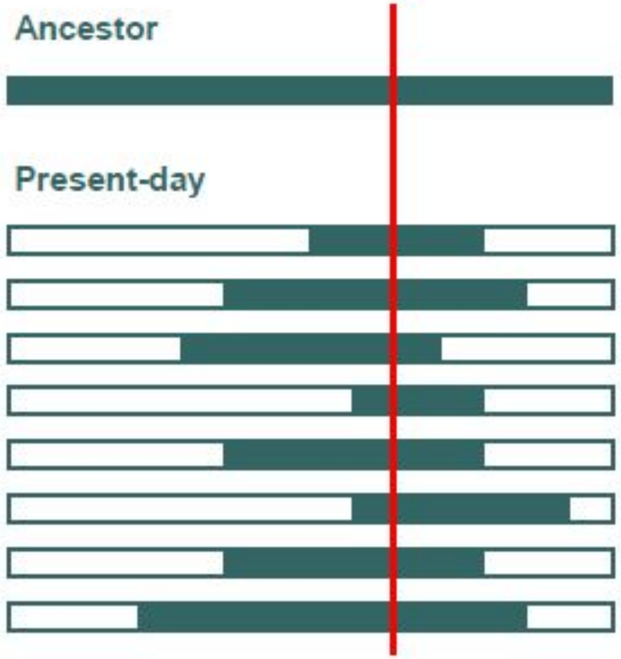
# One allele arose first, and then the other...



# Recombination generates new arrangements for ancestral alleles

# Linkage Disequilibrium

- Chromosomes are mosaics
- Extent and conservation of mosaic pieces depends on
  - Recombination rate
  - Mutation rate
  - Population size
  - Natural selection
- Combinations of alleles at very close markers reflect ancestral haplotypes



## Measuring linkage disequilibrium

- consider two loci (A and B), each segregating for two alleles ($A_1$, $A_2$, $B_1$, and $B_2$)
- there are four possible gametes with gene combinations (or haplotypes) present in the population:

| **Haplotype** | **Frequency** |
|---|---|
| $A_1B_1$ | $x_{11}$ |
| $A_1B_2$ | $x_{12}$ |
| $A_2B_1$ | $x_{21}$ |
| $A_2B_2$ | $x_{22}$ |

- allele frequencies can be expressed in terms of the gamete/haplotype frequencies:

| **Allele** | **Frequency** |
|---|---|
| $A_1$ | $p_1 = x_{11} + x_{12}$ |
| $A_2$ | $p_2 = x_{21} + x_{22}$ |
| $B_1$ | $q_1 = x_{11} + x_{21}$ |
| $B_2$ | $q_2 = x_{12} + x_{22}$ |

**- note that $p_1 + p_2 = 1$ and $q_1 + q_2 = 1$, and $\Sigma\ x_{ij} = 1.0$**

if alleles at the two loci are randomly associated with one another, then the frequencies of the four gametes are equal to the product of the frequencies of alleles it contains:

$$\mathbf{x_{11} = p_1q_1}$$
$$\mathbf{x_{12} = p_1q_2}$$
$$\mathbf{x_{21} = p_2q_1}$$
$$\mathbf{x_{22} = p_2q_2}$$

- in this situation, there is no linkage disequilibrium and gamete frequencies can be accurately followed using allele frequencies.
- if alleles at the two loci are not randomly associated, then there will a deviation (D) in the expected frequencies:

$$\mathbf{x_{11} = p_1q_1 + D}$$
$$\mathbf{x_{12} = p_1q_2 - D}$$
$$\mathbf{x_{21} = p_2q_1 - D}$$
$$\mathbf{x_{22} = p_2q_2 + D}$$

- this parameter **D is the coefficient of linkage disequilibrium first proposed by Lewontin and Kojima (1960).**
- the most common expression of D is:

$$\mathbf{D = x_{11}x_{22} - x_{12}x_{21}}$$

- where $x_{11}$ and $x_{22}$ are referred to as “coupling” gametes and $x_{12}$ and $x_{21}$ the “repulsion” gametes.
- D is thus the difference between these two gamete types.

**$D$ depends on allele frequencies**

Value can range from -0.25 to 0.25

Researchers suggested the value should be normalized

o Based on the theoretical maximum and minimum relative to the value of $D$

When D ≥ 0

$$D' = \frac{D}{D_{\max}}$$

$D_{max}$ is the smaller of $p_1q_2$ and $p_2q_1$.

When D < 0

$$D' = \frac{D}{D_{\min}}$$

$D_{min}$ is the larger of $-p_1q_1$ and $-p_2q_2$

**Another LD measure**

Correlation between a pair of loci is calculated using the following formula

o Value is *r*

o Or frequently $r^2$.

$$r^2 = \frac{D^2}{p_1 p_2 q_1 q_2}$$

**Ranges from**

o $r^2 = 0$

 Loci are in complete linkage equilibrium

o $r^2 = 1$

 Loci are in complete linkage disequilibrium

**Example:**
SNP locus A: A1=T, A2=C
SNP locus B: A1=1, A2=G
***Observed haplotype data***

| Haplotype | Symbol | Frequency |
|---|---|---|
| A1B1 | $x_{11}$ | 0.6 |
| A1B2 | $x_{12}$ | 0.1 |
| A2B1 | $x_{21}$ | 0.2 |
| A2B2 | $x_{22}$ | 0.1 |

***Calculated allelic frequency***

| Allele | Symbol | Frequency |
|---|---|---|
| A1 | *p1* | 0.7 |
| A2 | *p1* | 0.3 |
| B1 | *q1* | 0.8 |
| B2 | *q2* | 0.2 |

**$D = x_{11} – p_1q_1$; $D = 0.6 – (0.7)(0.8) = 0.6 – 0.56 = 0.04$**
**$D = (x_{11})(x_{22}) – (x_{12})(x_{21})$ $D = (0.6)(0.1) – (0.1)(0.2) = 0.04$**
***Calculating D'***
Since *D>0, use* $D_{max}$
$D_{max}$ *is* the smaller of *p*1*q*2 and *p*2*q*1
$p_1q_2 = 0.14$
$p_2q_1 = 0.24$

$$D' = \frac{D}{D_{\max}}; \qquad D' = \frac{0.04}{0.14} = 0.286$$

*Calculating $r^2$*

$$r^2 = \frac{D^2}{p_1p_2q_1q_2}$$

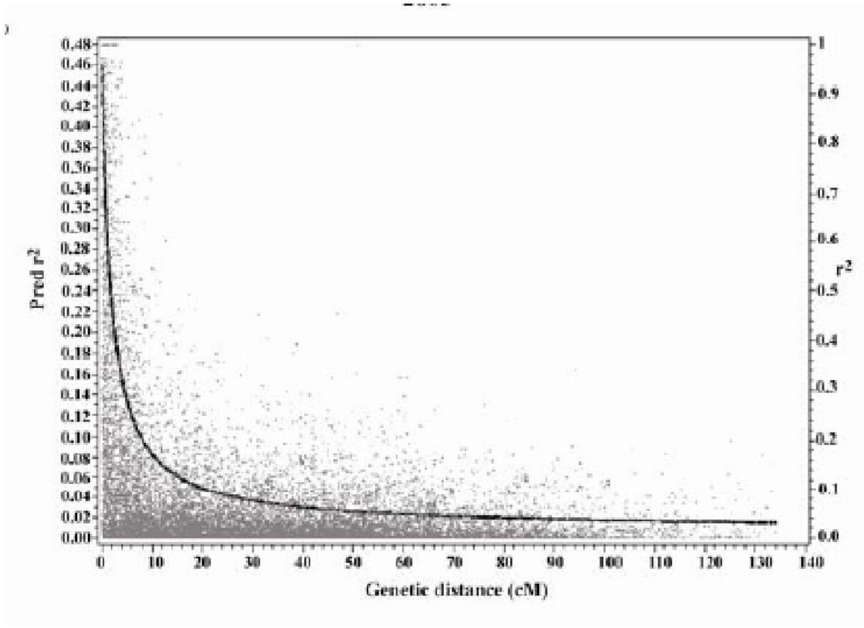
$$r^2 = \frac{0.04^2}{(0.7)(0.3)(0.8)(0.2)} = 0.048$$

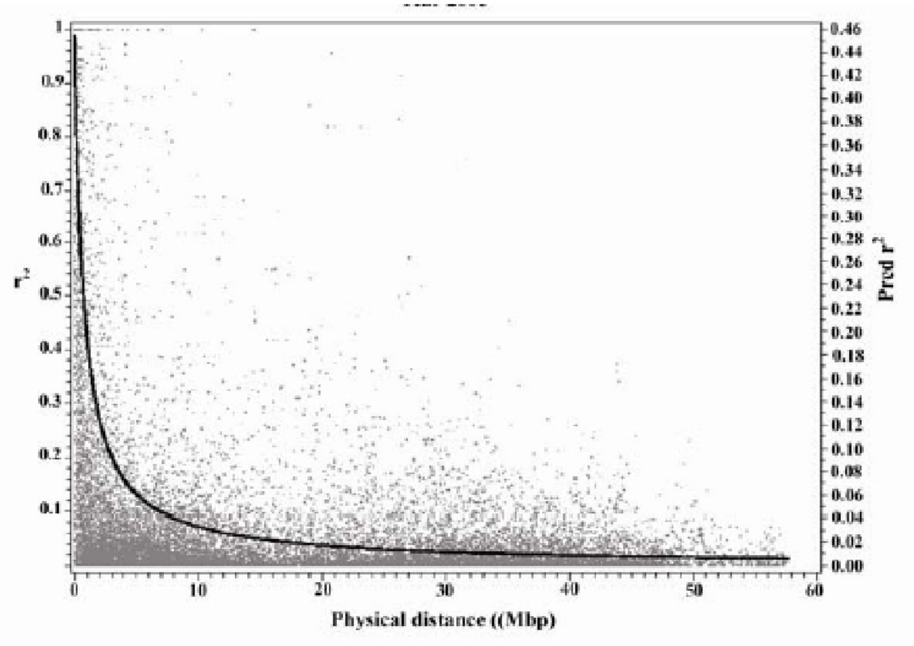
# LD decay

**Graphical relationship of**

All two loci pairwise comparisons

- $r^2$ relative to either genetic or physical distance
- $r^2$ vs. distance is calculated
  - Non-linear regression



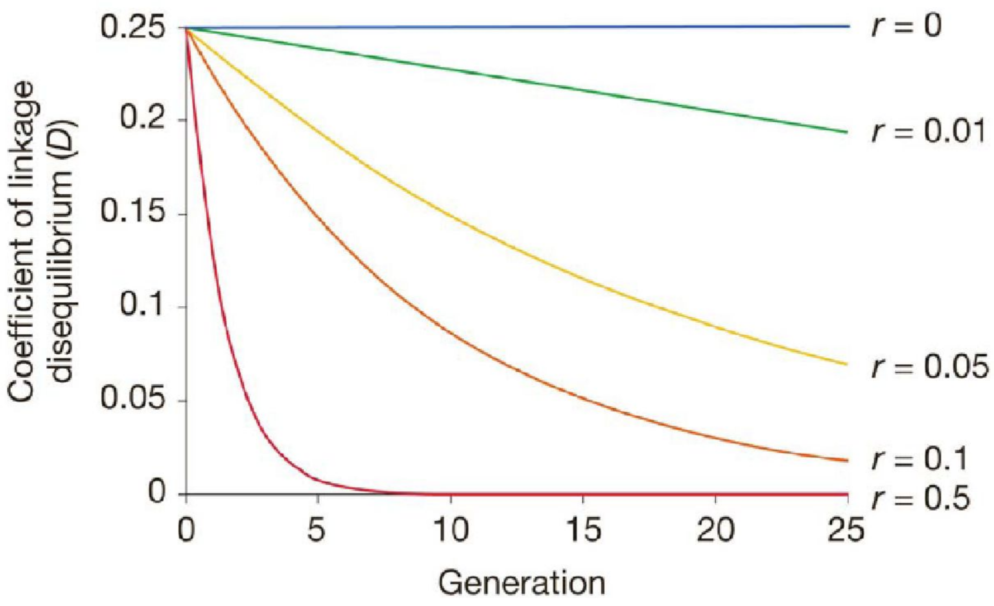




**What do the graphs tell us?**

- On average, how fast LD decays across the genome
- Useful to determine the number of markers needed for an association mapping experiment

# What Factors Affect Linkage Disequilibrium?

**Recombination**
- Changes arrangement of haplotypes
- Creates new haplotypes

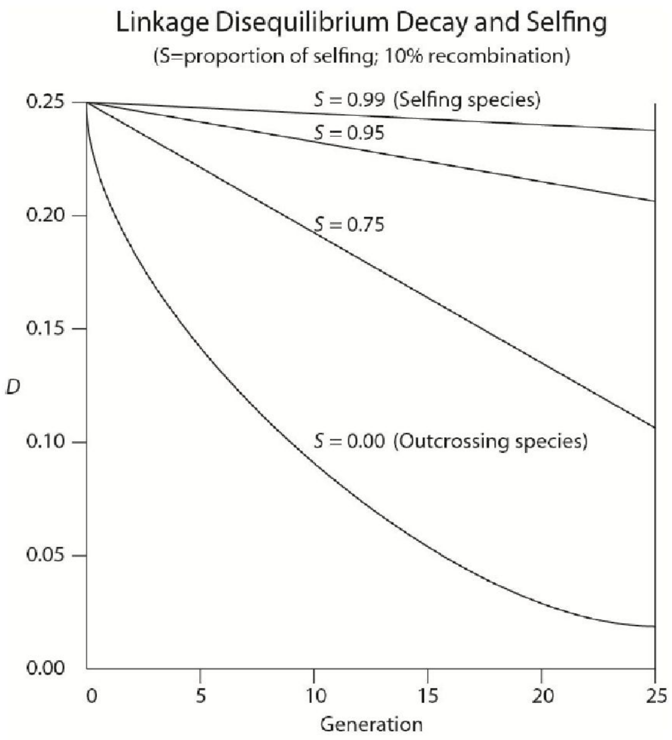



**Genetic Drift**
- Changes allele frequencies due to small population size

 LD changes depends on population size and recombination rate

- Smaller populations - New non-random associations appear - Larger LD values between some pairs of loci
- Larger populations - Less effect on LD

**Inbreeding**
- The decay of linkage disequilibrium is delayed in selfing populations
- Important for association mapping in self-pollinated crops




**Mutation**
 Effect is generally small absent recombination and gene flow
**Gene flow**
 LD becomes large if two populations intermating are genotypically distinct
 Not much of a problem if crossing between highly similar population found with most breeding programs

# LD decay in an outcrossed species